Question 1

Configure the MySQLscanner script to execute automatically every Wednesday at 3PM.

Approach

Use cron scheduler to run script weekly at specified day and time.

Commands

- `crontab -e` → edit cron jobs
- `0 15 * * 3` → schedule format (3 PM Wednesday)

Output

STEP 1 : open cron editor

```
┌──(aman㉿kali)-[~]
└─$ crontab -e
no crontab for aman - using an empty one
Select an editor.  To change later, run select-editor again.
  1. /bin/nano        <── easiest
  2. /usr/bin/vim.basic
  3. /usr/bin/vim.tiny

Choose 1-3 [1]: 1
```

STEP 2 : add scheduled task

```
  GNU nano 8.7.1
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
0 15 * * 3 /path/to/MySQLscanner.sh
```

```
┌──(aman㉿kali)-[~]
└─$ crontab -e
no crontab for aman - using an empty one
Select an editor.  To change later, run select-editor again.
  1. /bin/nano        <── easiest
  2. /usr/bin/vim.basic
  3. /usr/bin/vim.tiny

Choose 1-3 [1]: 1
crontab: installing new crontab
```

Key Observations

Cron uses 24-hour time format.
Day value 3 represents Wednesday.

What I learned

Cron enables precise automation of security scripts.

Question 2

Schedule the MySQLscanner script to execute on the 10th day of April, June, and August.

Approach

Define cron job specifying day of month and selected months.

Commands

- `crontab -e` → edit cron jobs
- `10 4,6,8` → specific months

Output

STEP 1 : open cron editor

```
┌──(aman㉿kali)-[~]
└─$ crontab -e
```

Schedule monthly execution

```
  GNU nano 8.7.1
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
0 0 10 4,6,8 * /path/to/MySQLscanner.sh
```

```
┌──(aman㉿kali)-[~]
└─$ crontab -e
crontab: installing new crontab
```

Key Observations

Multiple months separated by comma.
Runs once per selected month.

What I learned

Cron allows selective month scheduling.

Question 3

Configure the script to run at 10 AM every Tuesday, Wednesday, and Thursday.

Approach

Specify weekday range in cron expression.

Commands

- `crontab -e` → edit cron jobs
- `2-4` → Tuesday to Thursday

Output

STEP 1 : open cron editor

```
┌──(aman㉿kali)-[~]
└─$ crontab -e
```

STEP 2 : add weekday schedule

```
0 15 * * 3 /path/to/MySQLscanner.sh
0 0 10 4,6,8 * /path/to/MySQLscanner.sh
0 10 * * 2-4 /path/to/MySQLscanner.sh
```

Key Observations

Weekday numbering starts from Sunday=0.
Range simplifies multi-day scheduling.

What I learned

Cron ranges reduce repetitive entries.

Question 4

Configure the script to execute every day at 12 PM using cron shortcut syntax.

Approach

Use cron predefined expression for daily execution at noon.

Commands

- `@daily` → cron shortcut
- `crontab -e` → edit cron jobs

Output

STEP 1 : open cron editor

```
┌──(aman㉿kali)-[~]
└─$ crontab -e
```

STEP 2 : add daily schedule

```
0 15 * * 3 /path/to/MySQLscanner.sh
0 0 10 4,6,8 * /path/to/MySQLscanner.sh
0 10 * * 2-4 /path/to/MySQLscanner.sh
0 12 * * * /path/to/MySQLscanner.sh
```

Key Observations

Shortcut expressions simplify cron syntax.
Noon equals hour value 12.

What I learned

Cron shortcuts speed up configuration.

Question 5

Configure PostgreSQL service to automatically start when system boots.

Approach

Use update-rc.d to enable service startup during boot process.

Commands

- `update-rc.d` → manage startup services
- `systemctl enable` → alternative method

Output

STEP 1 : enable PostgreSQL at boot

```
┌──(aman㉿kali)-[~]
└─$ sudo update-rc.d postgresql enable
[sudo] password for aman:
```

Key Observations

Service persistence ensures availability after reboot.
Startup services may expose network ports.

What I learned

Boot persistence is critical for long-term system behavior.

Question 6

Install rcconf tool and configure PostgreSQL and MySQL services to start automatically.

Approach

Install rcconf package and enable database services via interface.

Commands

- `apt install rcconf` → install tool
- `rcconf` → manage startup services

Output

STEP 1 : install rcconf

```
┌──(aman㉿kali)-[~]
└─$ sudo apt install sysv-rc-conf
The following packages were automatically installed and are no longer required:
  curlftpfs       libavformat61           libfuse2t64             libmpeg2encpp-2.1-0t64  libplacebo351     librubberband2  libsphinxbase3t64  mesa-vdpau-drivers  python3-aiomcache       python3-wapiti-arsenic  vdpau-driver-all
  libaudio2       libconfig-inifiles-perl libgav1-1               libmplex2-2.1-0t64      libpocketsphinx3  libsframe2      libswscale8        pocketsphinx-en-us  python3-fs              python3-yaswfp
  libavfilter10   libcrypt-dev            libmjpegutils-2.1-0t64  libmupdf25.1            libpostproc58     libsnmp40t64    libvdpau-va-gl1    python3-aiocache    python3-terminaltables  ruby-unf-ext
Use 'sudo apt autoremove' to remove them.

Installing:
  sysv-rc-conf

Installing dependencies:
  libcurses-perl  libcurses-ui-perl  liblist-utilsby-perl

Summary:
  Upgrading: 0, Installing: 4, Removing: 0, Not Upgrading: 565
  Download size: 323 kB
  Space needed: 1,193 kB / 10.5 GB available

Continue? [Y/n] y
Get:1 http://kali.download/kali kali-rolling/main amd64 libcurses-perl amd64 1.46-1 [79.0 kB]
Get:4 http://mirrors.esto.network/kali kali-rolling/main amd64 sysv-rc-conf all 1.4.0-1 [43.1 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 libcurses-ui-perl all 0.9609-2 [186 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 liblist-utilsby-perl all 0.12-2 [15.5 kB]
Fetched 323 kB in 3s (119 kB/s)
Selecting previously unselected package libcurses-perl.
(Reading database… 439017 files and directories currently installed.)
Preparing to unpack …/libcurses-perl_1.46-1_amd64.deb…
Unpacking libcurses-perl (1.46-1)…
Selecting previously unselected package libcurses-ui-perl.
Preparing to unpack …/libcurses-ui-perl_0.9609-2_all.deb…
Unpacking libcurses-ui-perl (0.9609-2)…
Selecting previously unselected package liblist-utilsby-perl.
Preparing to unpack …/liblist-utilsby-perl_0.12-2_all.deb…
Unpacking liblist-utilsby-perl (0.12-2)…
Selecting previously unselected package sysv-rc-conf.
Preparing to unpack …/sysv-rc-conf_1.4.0-1_all.deb…
Unpacking sysv-rc-conf (1.4.0-1)…
Setting up liblist-utilsby-perl (0.12-2)…
Setting up libcurses-perl (1.46-1)…
Setting up libcurses-ui-perl (0.9609-2)…
Setting up sysv-rc-conf (1.4.0-1)…
update-alternatives: using /usr/sbin/sysv-rc-conf to provide /usr/sbin/chkconfig (chkconfig) in auto mode
Processing triggers for desktop-file-utils (0.28-1)…
Processing triggers for hicolor-icon-theme (0.18-2)…
Processing triggers for man-db (2.13.1-1)…
Processing triggers for kali-menu (2026.1.3)…
```

STEP 2 : run rcconf

```
┌──(aman㉿kali)-[~]
└─$ sudo sysv-rc-conf
Use of uninitialized value $state in numeric eq (==) at /usr/share/perl5/App/SysVRcConf/LSB.pm line 101, <$fh> line 14.
Use of uninitialized value $state in numeric eq (==) at /usr/share/perl5/App/SysVRcConf/LSB.pm line 101, <$fh> line 14.
Use of uninitialized value $state in numeric eq (==) at /usr/share/perl5/App/SysVRcConf/LSB.pm line 101, <$fh> line 15.
Use of uninitialized value $state in numeric eq (==) at /usr/share/perl5/App/SysVRcConf/LSB.pm line 101, <$fh> line 15.
Use of uninitialized value $state in numeric eq (==) at /usr/share/perl5/App/SysVRcConf/LSB.pm line 101, <$fh> line 16.
Use of uninitialized value $state in numeric eq (==) at /usr/share/perl5/App/SysVRcConf/LSB.pm line 101, <$fh> line 16.
Use of uninitialized value $state in numeric eq (==) at /usr/share/perl5/App/SysVRcConf/LSB.pm line 101, <$fh> line 17.
Use of uninitialized value $state in numeric eq (==) at /usr/share/perl5/App/SysVRcConf/LSB.pm line 101, <$fh> line 17.
Use of uninitialized value $state in numeric eq (==) at /usr/share/perl5/App/SysVRcConf/LSB.pm line 101, <$fh> line 18.
Use of uninitialized value $state in numeric eq (==) at /usr/share/perl5/App/SysVRcConf/LSB.pm line 101, <$fh> line 18.
Use of uninitialized value $state in numeric eq (==) at /usr/share/perl5/App/SysVRcConf/LSB.pm line 101, <$fh> line 19.
Use of uninitialized value $state in numeric eq (==) at /usr/share/perl5/App/SysVRcConf/LSB.pm line 101, <$fh> line 19.
Use of uninitialized value $state in numeric eq (==) at /usr/share/perl5/App/SysVRcConf/LSB.pm line 101, <$fh> line 31.
Use of uninitialized value $state in numeric eq (==) at /usr/share/perl5/App/SysVRcConf/LSB.pm line 101, <$fh> line 31.
Use of uninitialized value $state in numeric eq (==) at /usr/share/perl5/App/SysVRcConf/LSB.pm line 117, <$fh> line 152.
```

View image

```
 SysV Runlevel Config   +/-: start/stop   i: info   h: help   q: quit

 service      page 1/2   1     2     3     4     5     0     6     S
 ──────────────────────────────────────────────────────────────────────
 apache-htcacheclean    [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [~]    # systemd-managed
 apache2                [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [~]    # systemd-managed
 apparmor               [~]   [~]   [~]   [~]   [~]   [~]   [~]   [X]      systemd-managed
 atftpd                 [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [~]    # systemd-managed
 bluetooth              [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [~]    # systemd-managed
 cron                   [~]   [X]   [X]   [X]   [X]   [~]   [~]   [~]      systemd-managed
 cryptdisks             [~]   [~]   [~]   [~]   [~]   [ ]   [ ]   [X]      systemd-managed
 cryptdisks-early       [~]   [~]   [~]   [~]   [~]   [ ]   [ ]   [X]      systemd-managed
 dbus                   [~]   [X]   [X]   [X]   [X]   [~]   [~]   [~]      systemd-managed
 inetsim                [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [~]    # systemd-managed
 iodined                [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [~]    # systemd-managed
 ipsec                  [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [~]    # systemd-managed
 lightdm                [ ]   [X]   [X]   [X]   [X]   [ ]   [ ]   [~]      systemd-managed
 lm-sensors             [~]   [~]   [~]   [~]   [~]   [~]   [~]   [X]      systemd-managed
 mariadb                [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [~]    # systemd-managed
 miredo                 [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [~]    # systemd-managed
 mosquitto              [ ]   [X]   [X]   [X]   [X]   [ ]   [ ]   [~]      systemd-managed
 networking             [~]   [~]   [~]   [~]   [~]   [ ]   [ ]   [X]      systemd-managed
 nfs-common             [ ]   [~]   [~]   [~]   [~]   [ ]   [ ]   [ ]    # systemd-managed
 nginx                  [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [~]    # systemd-managed
 nmbd                   [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [~]    # systemd-managed
 openvpn                [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [~]    # systemd-managed
 pcscd                  [ ]   [X]   [X]   [X]   [X]   [ ]   [ ]   [~]      systemd-managed
 plymouth               [~]   [X]   [X]   [X]   [X]   [ ]   [ ]   [~]      systemd-managed
 plymouth-log           [~]   [~]   [~]   [~]   [~]   [~]   [~]   [X]      systemd-managed
 postgresql             [ ]   [X]   [X]   [X]   [X]   [ ]   [ ]   [~]      systemd-managed
 procps                 [~]   [~]   [~]   [~]   [~]   [~]   [~]   [X]      systemd-managed
 ptunnel                [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [~]    # systemd-managed
 redis-server           [ ]   [X]   [X]   [X]   [X]   [ ]   [ ]   [~]      systemd-managed
 redsocks               [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [~]    # systemd-managed
 rpcbind                [~]   [~]   [~]   [~]   [~]   [ ]   [ ]   [ ]    # systemd-managed
 rsync                  [~]   [X]   [X]   [X]   [X]   [~]   [~]   [~]      systemd-managed
 saned                  [ ]   [X]   [X]   [X]   [X]   [ ]   [ ]   [~]      systemd-managed
 screen-cleanup         [~]   [~]   [~]   [~]   [~]   [~]   [~]   [X]      systemd-managed
 smartmontools          [ ]   [X]   [X]   [X]   [X]   [ ]   [ ]   [~]      systemd-managed
 smbd                   [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [~]    # systemd-managed
 snmpd                  [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [~]    # systemd-managed
 speech-dispatcher      [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [~]    # systemd-managed
 ssh                    [~]   [ ]   [ ]   [ ]   [ ]   [~]   [~]   [~]    # systemd-managed
 sslh                   [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [~]    # systemd-managed
 stunnel4               [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [ ]   [~]    # disabled
 sudo                   [~]   [X]   [X]   [X]   [X]   [~]   [~]   [~]      systemd-managed

 <arrows>  m: toggle multiuser   /:find   d: defaults  r/BS: restore all/one
 0: stop [ ]  1: start [X]  2: floating [~]  space: toggle  </>: cycle state
```

Key Observations

GUI simplifies service management.
Database services often enabled at boot.

What I learned

Service startup configuration affects attack surface.

Security Insights

1. Scheduled scripts allow persistent automated reconnaissance.
2. Time-based execution helps evade monitoring patterns.
3. Multi-day scheduling supports repeated data collection.
4. Daily automation ensures continuous system interaction.
5. Boot services can maintain persistence after reboot.
6. Startup services increase exposure of network databases.